Modern C++20 Concepts, Ranges & Move Semantics

C++ latest
VERIFIED RUNTIME

KEY SYNTAX & IDIOMS

Core Syntax Declaration
```
function execute() { return true; }
```
Standard function signature

Assertion Verification
```
assert(result === expected);
```
Validates post-conditions

Error Handling
```
try { ... } catch (e) { ... }
```
Graceful failure recovery

Scoped Constants
```
const CONFIG = { mode: 'sandbox' };
```
Immutable reference declaration

COMMON PITFALLS & FIXES

Pitfall: Missing type assertions
Rule: Enforce strict compiler lint checks and compiler warnings

Pitfall: Uncontained side effects
Rule: Run untrusted student snippets in isolated container sandboxes

Pitfall: Unbounded recursion
Rule: Set configurable 20s timeouts to contain infinite loops

TINY VERIFIED EXAMPLE: VERIFIED EXECUTION

SOURCE CODE:
```
console.log('Verified Course Material Studio');
```

REAL CAPTURED OUTPUT:
```
Verified Course Material Studio
```